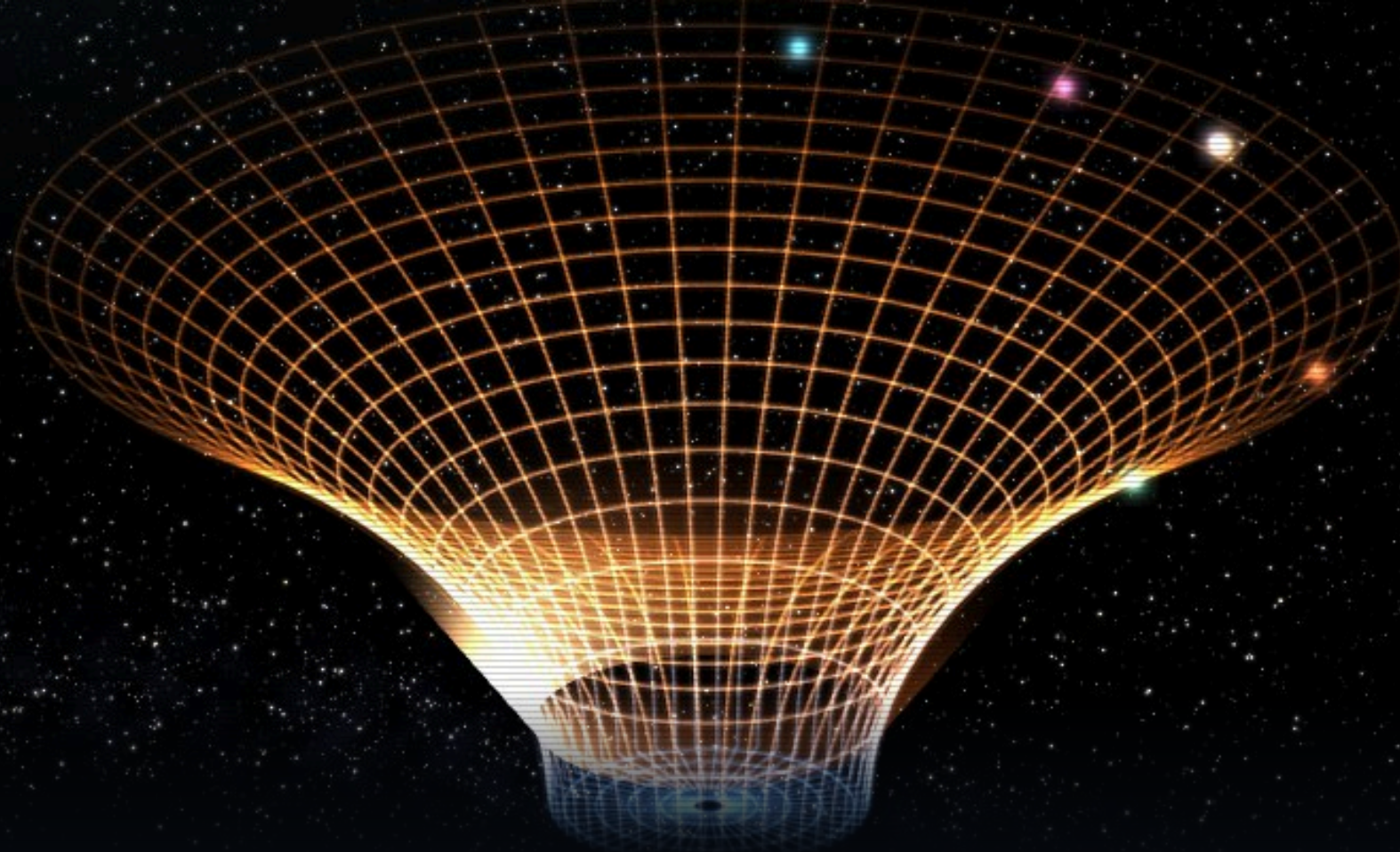




BROKILD • VST3 + STANDALONE • FLIGHT MANUAL

GRAVITY WELL

A bassline synthesiser built on one idea: mass curves time. Turn it up and every movement in the sound slows, deepens and finally stops at the horizon — while the note stays exactly in tune.

MASS CURVES TIME

Near a heavy object, clocks run slow. A clock at distance r from a black hole ticks at

$$\text{rate} = \text{sqrt}(1 - r_s / r)$$

of its normal speed, where r_s , the Schwarzschild radius, grows with the mass.

Gravity Well uses that exact formula as its modulation system. Every wobble in the sound — each LFO, the filter and third envelopes, the random source — is a clock sitting at a **RADIUS** you choose. Turn up **MASS** and the clocks near the hole slow down while the far ones barely notice.

Because the factor bends steeply near the horizon, the **ratios** between modulators change as the mass rises. A shared “mod speed” knob makes the same patch slower; this makes a *different* patch — the movement re-orders itself.

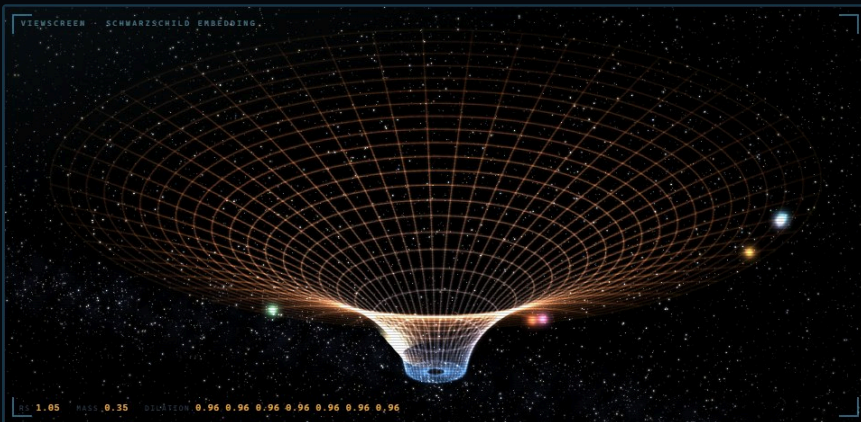
And at the horizon the formula reaches zero. A modulator that falls inside it **freezes** and holds its last value: a wobble stopped mid-swing at the bottom of a heavy note, which picks up again from where it stopped when you let go.

WHY IT DOES NOT GO OUT OF TUNE

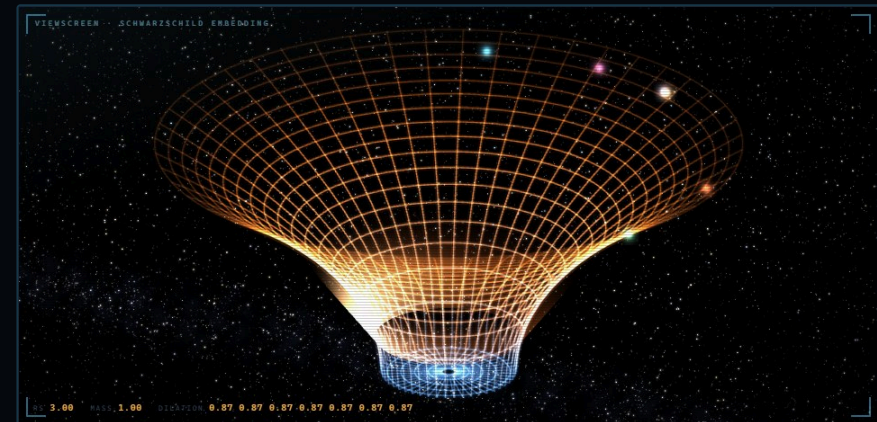
Light climbing out of a gravity well loses energy and arrives redder — lower in frequency. Applied to a bass note that would pull every note flat. So the **redshift** is applied to the **colour** of the sound, never its pitch: the filters and the vowel bank sink, the fundamental stays where you played it. The sound gets darker and heavier and stays in tune.

WHERE THE MASS COMES FROM

MASS is three things added together: the knob, how loud the voice is playing, and how hard you struck the key. Play harder and the well deepens by itself; back off and it lifts. At MASS 0 the well is flat and Gravity Well is an ordinary, very good bass synth.



Light. A shallow well: the clocks run nearly free.



Heavy. The funnel deep, the horizon ring bright at the throat, every clock at 0.87.

THE PANEL AT A GLANCE



REESE held, MASS at 72 %: the well deep, every clock slowed to 0.87 of its speed.

THE VIEWSCREEN

The well is Flamm's paraboloid, the real embedding diagram of the Schwarzschild metric. Its depth is the mass the engine is using right now, and the coloured lights are the modulators at their own radii. When a light stops on screen, it has stopped in the sound. Drag to orbit, wheel for elevation.

THE READOUT

RS the horizon radius, MASS the live mass, DILATION each modulator's clock speed (1.00 = free, 0.00 = frozen).

THE PILLARS

Left: filter A's cutoff, resonance and circuit, always in reach. Right: the six macros.

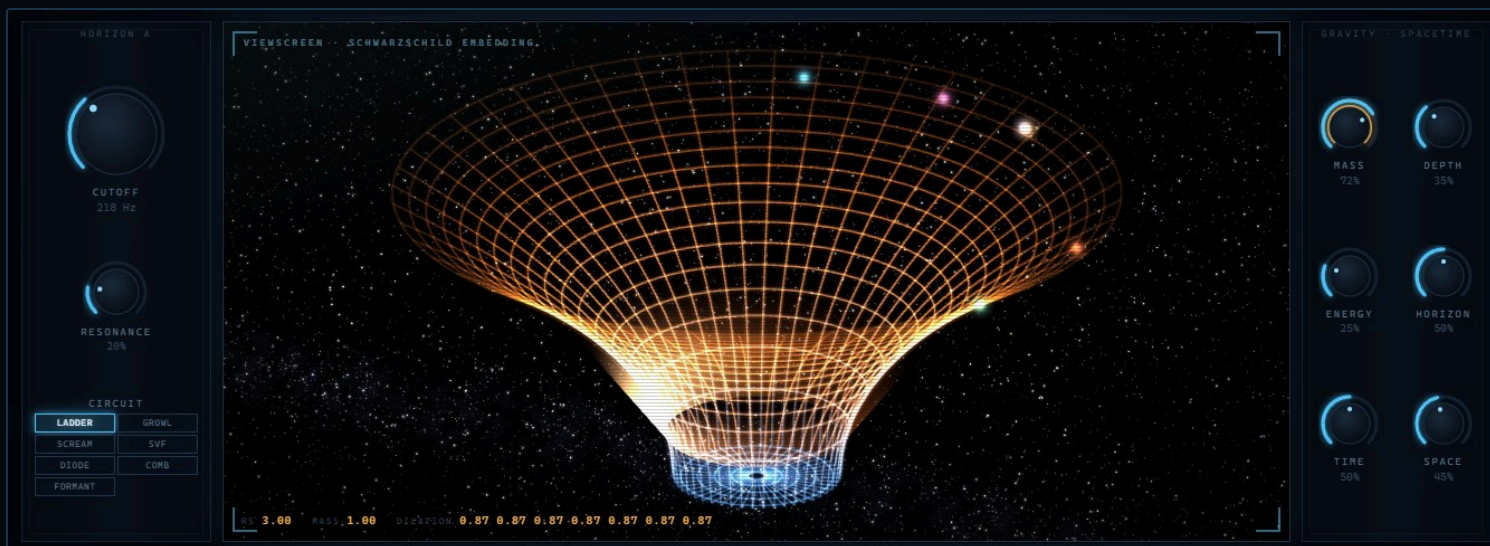
THE DESK

Seven consoles — HELM, REACTOR, HORIZONS, CHRONO, ORBITS, MATRIX, OUTPUT — plus WORLD FX and SETTINGS. Each is flanked by line displays drawn from the engine.

THE HEADER

The patch window (click the name for the chart), SAVE, and PANIC. KEYS opens a four-octave keyboard in the foot.

MOST OF THE WAY IN SIX KNOBS



MASS

The weight of the hole. Deepens the well, slows the clocks, sinks the colour. Past noon it acts on the sound directly: the top darkens and the sub gathers.

DEPTH

Opens the **singularity**: a protected low core that goes round the filters and drive, so the bottom survives whatever you do to the rest.

ENERGY

Drives the chain harder. It adds to the drive amount on OUTPUT, and every drive is level-matched, so it adds grind, not loudness.

HORIZON

How large the hole can grow for a given mass, and a tilt on both filter cutoffs together: down is darker and more captured, up is brighter and freer.

TIME

The master clock for every free-running LFO and the random source, before gravity slows them. Noon is neutral; one side quarters the speed, the other quadruples it. Tempo-synced LFOs ignore it.

SPACE

Stereo width: how far the accretion disk's voices spread across the field.

THE QUICK CONSOLE



HELM gathers the controls you reach for most, each marked with the console it lives on. The left display is the filter response you are hearing.

GRAVITY

TRACK	How much the playing level adds to the mass. High: loud notes weigh more.
VELOCITY	How much the key strike adds to the mass.
REDSHIFT	How far the colour sinks with the mass — every filter circuit, and the vowel bank most of all. Never the pitch.
RINGDOWN	The body. A black hole that is disturbed rings at its own inharmonic tones, and more mass rings lower and longer. This is that bank of tones, struck by each note.
RING DECAY / TONE	How long the body rings, and how bright it is.

THE REST OF THE HELM

SOURCE is oscillator A’s shape and level and the sub. **HORIZON A** is the first filter’s mode, envelope depth, drive and key follow. **DRIVE**, **AMP** and **OUT** mirror their consoles.

Everything on the helm is the same control as on its home console, not a copy: move it in one place and it moves in the other.

FIRST FIVE MINUTES

Pick **SCHWARZSCHILD**, hold a low note and turn **MASS** slowly up. Watch the lights slow on the screen and listen to the movement thicken. Past noon, hear the top sink into the sub.

THE SOURCES



SWARM: oscillator A on the wavetable engine, eight disk voices on Keplerian orbits. The left display draws one cycle of each oscillator.

OSCILLATORS A AND B

ANALOGUE is band-limited saw, pulse and triangle morphed by **SHAPE**, with **WIDTH** for the pulse.

WAVETABLE reads eight precomputed tables — **BASIC**, **OCTAVE**, **FORMANT**, **METAL**, **GLASS**, **GROWL**, **FOLD**, **SWARM** — clean at every pitch. **OCTAVE** spans ± 3 , **SEMI** and **FINE** tune.

CROSS

SYNC hard-resets B from A for the tearing sync sweep.

PHASE MOD lets A bend B's phase — bright and stable at any sample rate. **RING MOD** multiplies them for metal.

SUB AND NOISE

The **sub** sits one or two octaves below oscillator A, sine or square, and stays in phase with it so it reinforces rather than cancels. **NOISE** with **COLOUR** from dark to bright.

ACCRETION DISK

Unison, but not a flat detune: up to eight voices orbit at Keplerian radii, where the inner ones run faster than the outer. **SPREAD** is how far out the disk reaches, **ORBIT** how fast it turns, **WIDTH** how wide it sits in stereo. The shear between orbits is the beating.

COST

Wavetables are read from ready-made tables, so even an eight-voice swarm costs a few percent of a core. The heaviest factory patch is measured on every build.

ENGINES PER OSCILLATOR

A and B choose their engine independently, so an analogue body under a wavetable top is one click each.

TWO FILTERS, SEVEN CIRCUITS



EVENT HORIZON: a vowel bank into a state-variable high-pass, in series. The right display is the signal path as it is wired now.

LADDER	The warm four-pole classic. Resonance thins the bass, as it should.
GROWL	A Sallen-Key with diodes in the feedback: the aggressive MS-20 bite.
SCREAM	The same circuit pushed hot; it sings from noon and can be played on its resonance.
SVF	Clean state-variable, low, band, high-pass and notch. The high-pass keeps mud out.
DIODE	The 303 ladder: 18 dB/oct, and its resonance keeps the bass (+2.3 dB at 40 Hz where a ladder loses 17). The diodes clip unevenly, so it adds even harmonics.
COMB	A tuned comb: metallic, hollow, and tunable to the note.
FORMANT	A vowel bank that morphs u-o-a-e-i with no jumps. Talking bass, growl.

ROUTING

SERIES A into B. **PARALLEL** both at once, blended by A/B. **SPLIT** sends everything below the SPLIT frequency to A and above to B — wreck the mids, keep the floor.

PER FILTER

CUTOFF, RESONANCE and DRIVE; KEY follow; ENV is how far the filter envelope opens it (also on CHRONO). Modes that a circuit does not have are dimmed.

SINGULARITY

A low band below FREQ taken around the whole filter and drive chain and added back at the end, as the filtered low of the source or a pure sine. LEVEL here, or the DEPTH macro. The bottom nothing can destroy.

ENVELOPES, AND THE ORDER OF THINGS



TECHNO PLUCK, struck five times. Left: the three envelope shapes. Right: what each envelope actually did, live.

AMP ENVELOPE

Level only. It sits after the filters and the drive, so a loud strike and a quiet sustain have the same tone — the amp shapes the loudness and nothing else.

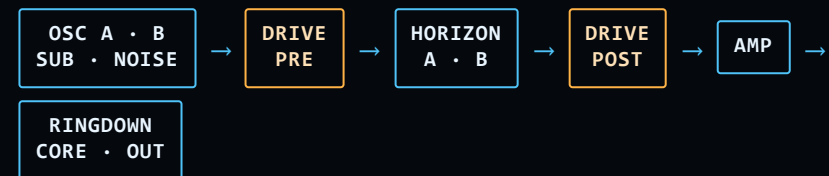
FILTER ENVELOPE

Opens the filters by the amounts in **DEPTH** (TO A CUT, TO B CUT). Every note in DUO has its own. Every factory patch has it wired, even where it only holds the filter still.

ENV 3 • FREE

Routes anywhere through the MATRIX. CURVE bends each stage; RADIUS puts the filter and third envelopes into the well, so gravity stretches them too.

THE SIGNAL CHAIN



The drive sits in **one** of its two places, chosen on OUTPUT. **PRE** (the default) distorts the raw oscillators and lets the filters shape what it made, as most hardware does. **POST** distorts the filtered sound, which is rougher and more obvious.

The singularity core bypasses everything between the source and the output, then joins it at the end.

THE CLOCKS IN THE WELL



WOBBLE: LFO 1 on a tempo division. The right display is every modulator's output, live — a clock inside the horizon draws a flat line and reads FROZEN.

LFO 1–3

RATE, and SHAPE: sine, triangle, saw, square or **chirp**. SYNC locks it to the host tempo in straight, dotted and triplet divisions, aligned to the bar while the transport runs.

RADIUS

Where the clock sits in the well. At 100 % it is far out and runs free; bring it in and gravity slows it as the mass rises. The light on the screen wears the knob's colour.

TARGET AND AMOUNT

Each LFO can drive one destination directly: pick it under TARGET, set AMT (either way). It shows in the MATRIX alongside the eight slots, and leaves them free.

RANDOM

A drifting random value, RATE and SMOOTH — from stepped to a slow wander. It has a radius too.

CHIRP

The inspiral: two masses spiralling together sweep up faster and faster, $f \propto (t_c - t)^{-3/8}$. TIME is how long the sweep takes. A shape, not a button.

THE TIME MACRO

Every free-running clock and the random source run at TIME's speed before gravity slows them. Noon changes nothing.

EIGHT ROUTES, AND THE ONES BUILT IN



WOBBLE: slot 1 sends LFO 1 to A CUTOFF. The dashed curves are the built-in routes — the amp envelope to LEVEL and the filter envelope to the cutoff.

A SLOT

SOURCE, DEST and AMOUNT. Amount is bipolar: negative runs the route backwards.

SOURCES

ENV1–3, LFO1–3, RANDOM, CHIRP, MASS, VELOCITY, KEY, WHEEL and PRESSURE. MASS as a source means the weight of the note can move anything.

DESTINATIONS

Pitch, both shapes, width, phase mod, ring, sub, noise, both cutoffs and resonances, split, A/B blend, disk, drive, mass, redshift, ringdown, pan and level.

THE NETWORK

Solid lines are the slots and the ORBITS targets; dashed lines are what is always wired. Each line breathes with its source, so you can see which routes are doing the work.

MASS ON MASS

Route an LFO to **MASS** and the hole itself breathes: the clocks near the horizon slow and speed up with it, and a slow cycle freezes them at the bottom of every swing.

DRIVE, VOICE, LEVEL



DOOM WEIGHT: PLASMA after the filters. The right display is the output level over time, amber where the ceiling is shaping the peaks.

FOUR DRIVES

IDLE BURN	A gentle rounding: warmth and a little thickness.
ION	A symmetric saturator that hardens towards a clip: odd harmonics, edge.
PLASMA	Lopsided, so it adds even harmonics too: the thickest and dirtiest.
COLLAPSE	Turns from saturation into a wavefolder as AMOUNT rises: the wave folds back on itself.

Each drive is level-matched against its own measured make-up, so turning AMOUNT up changes the grind and not the loudness. **POSITION** puts it before (PRE) or after (POST) the filters.

VOICE

MONO or **DUO** (two notes, each with its own envelopes). **GLIDE**: overlapping notes slide into each other instead of retriggering, which is how acid lines are played from a piano roll. **BEND** is the pitch wheel's range, **TRANSPOSE** moves the whole instrument up to an octave either way, and **VELOCITY** sets how hard each note hits.

CHARACTER

VINTAGE adds drift and small tolerances that keep a held bass alive. **TILT** leans the whole tone darker or brighter.

LEVEL

TRIM is the patch's own level: the factory bank is levelled to within about a decibel of each other. A soft ceiling guards the output.

AROUND THE EDGES



WORLD FX, the rack every Brokild synth shares.

WORLD FX

Twelve effects you can switch on, reorder and blend — tube, phaser, chorus, tremolo, gate, grit, compressor, echo, space, shimmer, rotary, a step glitcher — with presets, a rack mix and five macros your host can automate. Saved with the patch.

PATCHES

Everything happens in the **patch chart**. 00–31 are the factory bank and cannot be overwritten; your own go in numbered slots 032–199, each a file in Documents\Brokild patches\Gravity Well, shared by every project. **SAVE** opens the chart with your slots in view: name the sound, click a slot to aim at it, **SAVE** — or **REPLACE**, in amber, if the slot is taken. **CLEAR SLOT** (two clicks) moves a patch to a Deleted folder, so nothing is ever lost by accident. The header arrows step through the bank and then your slots.

SETTINGS

PANEL BRIGHTNESS lifts or dims the whole ship; **HUD LINES** sets how bold the line displays are. Both belong to this computer, not to a patch.

THIRTY-TWO PATCHES, ORDINARY TO STRANGE

FOUNDATION

Honest bass. Mass low; the well mostly flat.

- 00 SCHWARZSCHILD
- 01 FLAT SPACE
- 02 SOLID BODY
- 03 ROUND DUB
- 04 TECHNO PLUCK
- 05 SQUARE ONE
- 06 SUB ONLY
- 07 SOFT ORBIT
- 08 SHORT FALL
- 09 SLOW MASS

WORKING

The genre tools: acid, reese, growl, wobble, doom.

- 10 ACID 303
- 11 ACID SCREAM
- 12 REESE
- 13 GROWL
- 14 WOBBLE
- 15 INDUSTRIAL
- 16 DOOM WEIGHT
- 17 PICK ATTACK
- 18 RUBBER
- 19 GLASS BASS

CHARACTER

One feature each, pushed to be heard.

- 20 COMB TUNED
- 21 FORMANT TALK
- 22 RING METAL
- 23 CHIRP DROP
- 24 SWARM
- 25 REDSHIFT
- 26 RINGDOWN
- 27 FROZEN CLOCK

EVENT HORIZON

The physics as the instrument. Deep mass, frozen clocks.

- 28 EVENT HORIZON
- 29 TIDAL
- 30 SINGULARITY
- 31 HAWKING

LEVELS

Every factory patch is measured on its loudest 200 ms and trimmed to the same level, so stepping through the bank does not jump.

PATCH CHART

FACTORY 00-31YOURS 032-199SAVE

DOOM WEIGHT

SLOT 036 FREE

SAVE

032 - 073 11 OF 168 SLOTS USED

CLICK A SLOT TO AIM AT IT • DOUBLE-CLICK AN EMPTY ONE TO SAVE THERE

032 DEEP ORBIT	039 -	046 -	053 PERIMELION	060 DOOM LOW	067 -
033 COLD START	040 TIDAL LOCK	047 DRONE FOR K	054 -	061 -	068 -
034 HEAVY DUB	041 SLOW FALL	048 -	055 -	062 -	069 -
035 ACID LINE 2	042 -	049 -	056 -	063 -	070 -
036 -	043 -	050 -	057 -	064 -	071 -
037 -	044 -	051 -	058 -	065 -	072 -
038 NIGHT BUS	045 -	052 WIDE REESE	059 -	066 -	073 -

Each slot is a file in Documents\Brokild patches\Gravity Well, shared by every project. To rename, save to the same slot. To move, save to the new slot and clear the old one. A cleared patch is kept in the Deleted folder there.

The patch chart, saving: slot 036 aimed at, in amber. Six columns of seven slots slide a column (◀ ▶, or the arrow keys) or a page (◀ ◀ ▶ ▶, PageUp/PageDown) at a time. Open it on your patches by clicking the patch name in the header.

GRAVITY WELL

Format	Windows VST3 instrument and a standalone application
Parameters	143 host parameters, plus five World FX macros
Sources	two oscillators (analogue or eight wavetables), sub, noise, sync, phase and ring mod, an eight-voice accretion disk
Filters	seven circuits in two banks: series, parallel or split
Modulation	three envelopes, three LFOs with tempo sync, random, chirp, eight matrix slots and three orbit targets
Drive	four level-matched drives, before or after the filters
Voices	mono or duo, with glide
Patches	32 factory, levelled; 168 numbered user slots (32–199), one file each, shared by every project
Panel	needs the Microsoft WebView2 runtime, present on almost every current Windows

INSTALLING

Copy the **Gravity Well.vst3** folder into Program Files\Common Files\VST3 and rescan in your host. **Gravity Well.exe** is the standalone and needs no installation.

THE PHYSICS, HONESTLY

The formulas are the real ones — Schwarzschild time dilation for the clocks, the embedding diagram for the picture, the published quasi-normal mode ratios for the ringdown, the inspiral law for the chirp. What they drive is sound, not a simulation: nothing here claims to be a black hole, only to borrow how one bends time.

CREDITS

Peter Bøggild, September 2026. The newest member of the Brokild Collection. Free. Source: github.com/peterboggild/BrokildApps